

ECEN-601: Homework 4

Due on October 21, 2025 at 11:59 pm

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Problem 1

Let

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -1 & 0 & 3 & 5 \\ 1 & -2 & 1 & 1 \end{bmatrix}.$$

Find a row-reduced echelon matrix R which is row-equivalent to A and an invertible 3×3 matrix P such that $R = PA$.

Solution

$$\left[\begin{array}{cccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 3 & 5 & 0 & 1 & 0 \\ 1 & -2 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 5 & 1 & 1 & 0 \\ 0 & -4 & 0 & 1 & -1 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 5 & 1 & 1 & 0 \\ 0 & 0 & 8 & 11 & -1 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 5 & 1 & 1 & 0 \\ 0 & 0 & 8 & 11 & 1 & 2 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 5/2 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1 & 11/8 & 1/8 & 2/8 & 1/8 \end{array} \right] \quad (\text{REF})$$

$$\left[\begin{array}{cccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 5/2 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1 & 11/8 & 1/8 & 2/8 & 1/8 \end{array} \right] \rightarrow \left[\begin{array}{cccc|ccc} 1 & 0 & 0 & -7/8 & 3/8 & -1/4 & 3/8 \\ 0 & 1 & 0 & -1/4 & 1/4 & 0 & -1/4 \\ 0 & 0 & 1 & 11/8 & 1/8 & 1/4 & 1/8 \end{array} \right] \quad (\text{RREF})$$

The final augmented matrix corresponds to the matrices $[R \mid P]$.

Problem 2

Let V be the set of all complex-valued functions f on the real line such that (for all $t \in \mathbb{R}$)

$$f(-t) = \overline{f(t)}.$$

The bar denotes complex conjugation. Show that V , with the operations

$$\begin{aligned}(f + g)(t) &= f(t) + g(t) \\ (cf)(t) &= cf(t)\end{aligned}$$

is a vector space over the field of real numbers. Give an example of a function in V which is not real-valued.

Solution

Starting with the first operation:

$$\begin{aligned}(f + g)(-t) &= f(-t) + g(-t) = \overline{f(t)} + \overline{g(t)} \\ &= \overline{f(t) + g(t)} \\ &= \overline{(f + g)(t)}\end{aligned}$$

Now the second operation:

$$\begin{aligned}(cf)(-t) &= cf(-t) = c\overline{f(t)} \\ &= \overline{cf(t)} = \overline{(cf)(t)}\end{aligned}$$

Problem 3

Let F be a field and let n be a positive integer ($n \geq 2$). Let V be the vector space of all $n \times n$ matrices over F . Which of the following sets of matrices A are subspaces of V ?

- (a) all invertible A
- (b) all non-invertible A
- (c) all A such that $AB = BA$, where B is some fixed matrix in V
- (d) all A such that $A^2 = A$

Solution

Part (a): The set of all invertible matrices A is not a subspace of V since $0A$ is not invertible.

Part (b): There exist non-invertible matrices whose sum is invertible. Therefore, the set of all non-invertible matrices A is not a subspace of V .

Part (c): Say we have two matrices A and A' are in the set such that $AB = BA$. Then,

$$(A + A')B = AB + A'B = BA + BA' = B(A + A').$$

Therefore, the set of all matrices A such that $AB = BA$ is a subspace of V .

Part (d): Here A and A' are in the set such that $A^2 = A$. Then,

$$(A + A')^2 = A^2 + AA' + A'A + A'^2 = A + AA' + A'A + A' \neq A + A'.$$

Therefore, this set of matrices is not a subspace of V .

Problem 4

Let W_1 and W_2 be subspaces of a vector space V such that the set-theoretic union of W_1 and W_2 is also a subspace. Prove that one of the spaces W_i is contained in the other.

Solution

The union of W_1 and W_2 , $W_1 \cup W_2$, is a subspace of a vector space V .

If $u \in W_1$ and $u \notin W_2$, and $v \in W_2$ and $v \notin W_1$, then the sum $u + v$ is necessarily in the union of W_1 and W_2 .

This contradicts the idea that $u \in W_1$ but $v \notin W_1$, and vice versa. Therefore, if u is in one subspace, it must also be in the other.

Thus, one of the spaces W_i must be contained in the other.

Problem 5

Let W_1 and W_2 be subspaces of a vector space V such that $W_1 + W_2 = V$ and $W_1 \cap W_2 = \{\underline{0}\}$. Prove that for each vector α in V there are unique vectors α_1 in W_1 and α_2 in W_2 such that $\alpha = \alpha_1 + \alpha_2$.

Note: If $S_1, S_2, \dots, S_k$ are subsets of a vector space V , the set of all sums

$$\alpha_1 + \alpha_2 + \dots + \alpha_k$$

of vectors α_i in S_i is called the sum of the subsets $S_1, S_2, \dots, S_k$ and is denoted by

$$S_1 + S_2 + \dots + S_k$$

or by

$$\sum_{i=1}^k S_i.$$

Solution

We have $\alpha_1 + \dots + \alpha_k \in W_1 + \dots + W_k$ and some $\alpha'_1 + \dots + \alpha'_k \in W_1 + \dots + W_k$.

If the components of α are unique, then

$$\alpha_1 + \alpha_2 = \alpha'_1 + \alpha'_2 \implies \underbrace{\alpha_1 - \alpha'_1}_{\in W_1} = \underbrace{\alpha'_2 - \alpha_2}_{\in W_2}.$$

This vector is in both W_1 and W_2 , so it must be the zero vector condition $W_1 \cap W_2 = \{\underline{0}\}$. Therefore, $\alpha_1 = \alpha'_1$ and $\alpha_2 = \alpha'_2$, which means there are unique vectors α_1 in W_1 and α_2 in W_2 such that $\alpha = \alpha_1 + \alpha_2$.

Problem 6

Let V be a vector space over the complex numbers. Suppose that α , β , and γ are linearly independent vectors in V . Prove that $(\alpha + \beta)$, $(\beta + \gamma)$, and $(\gamma + \alpha)$ are linearly independent.

Solution

If $(\alpha + \beta)$, $(\beta + \gamma)$, and $(\gamma + \alpha)$ are linearly independent, then

$$a_1(\alpha + \beta) + a_2(\beta + \gamma) + a_3(\gamma + \alpha) = 0$$

has only the trivial solution $a_1 = a_2 = a_3 = 0$.

Checking by row reduction with

$$a_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + a_3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

Equivalently,

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore the vectors $(\alpha + \beta)$, $(\beta + \gamma)$, and $(\gamma + \alpha)$ are linearly independent.